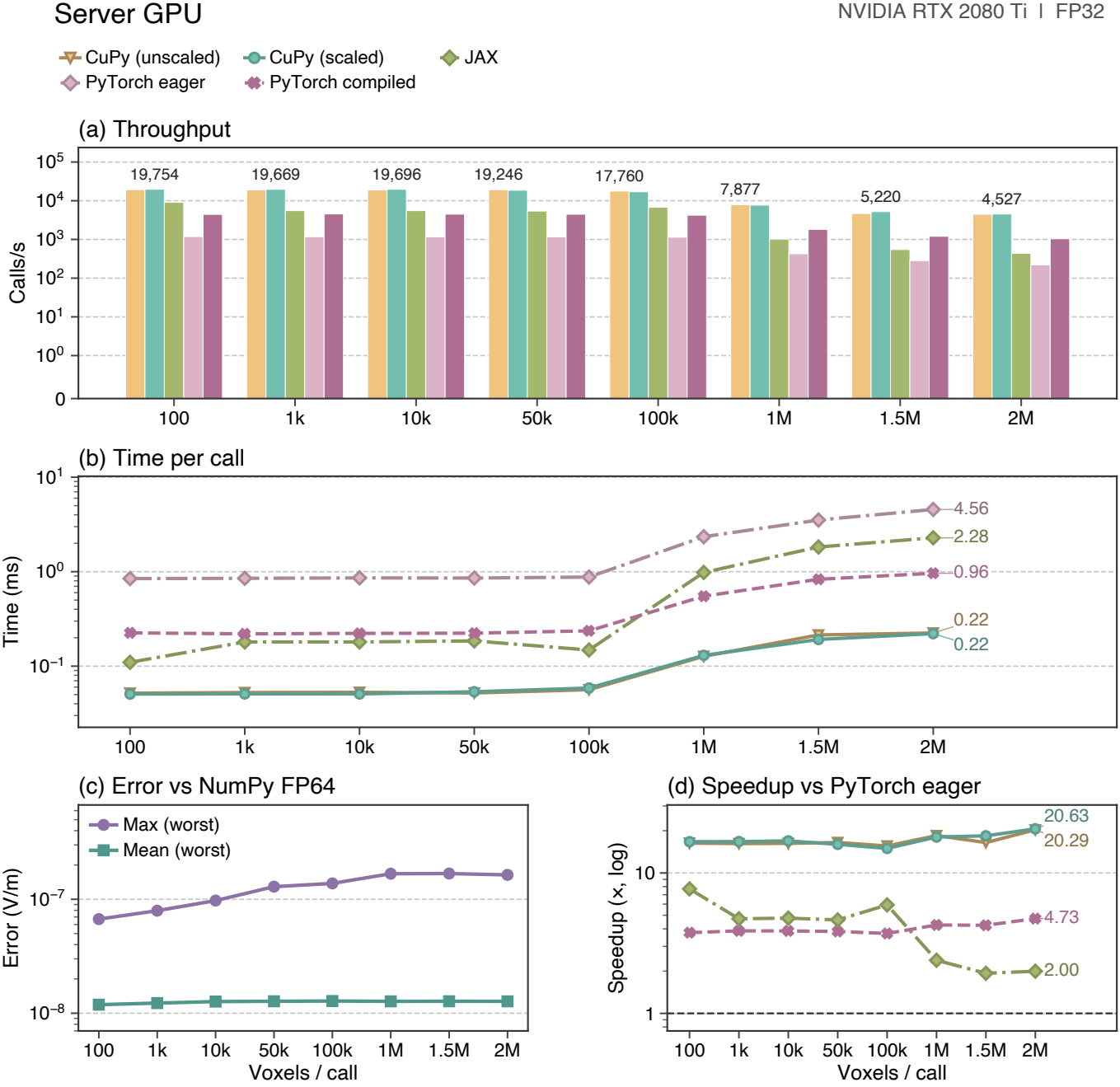
Server GPU
NVIDIA RTX 2080 Ti | FP32
CuPy (unscaled)
CuPy (scaled)
JAX
PyTorch eager
PyTorch compiled
(a) Throughput
Calls/s
19,754
19,669
19,696
19,246
17,760
7,877
5,220
4,527
100
1k
10k
50k
100k
1M
1.5M
2M
(b) Time per call
Time (ms)
4.56
2.28
0.96
0.22
0.22
(c) Error vs NumPy FP64
Max (worst)
Mean (worst)
Error (V/m)
Voxels / call
(d) Speedup vs PyTorch eager
Speedup (×, log)
20.63
20.29
4.73
2.00
Voxels / call